

AI Steam Game Advisor

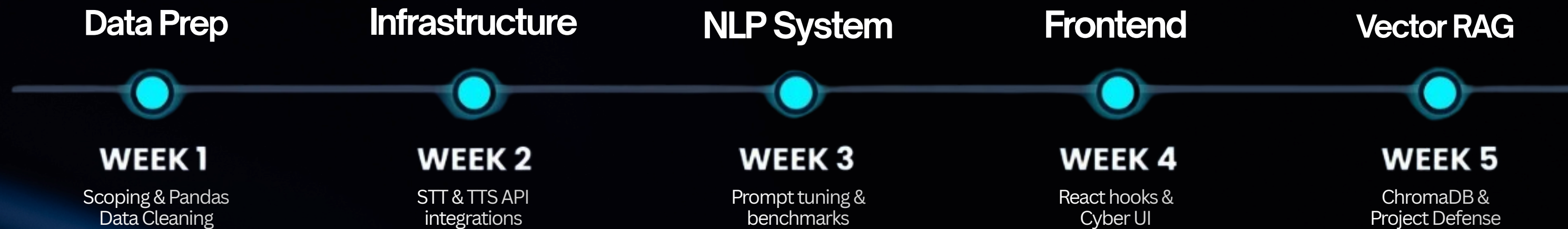
 Track : AI Avatars

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Our 5 Week Route



ISSAI

NAZARBAYEV UNIVERSITY | Institute of Smart Systems and Artificial Intelligence

Week 1: Fundamentals of Data & Infrastructure

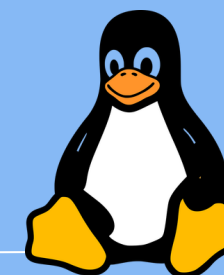
Python & Data Handling



Built loops and structures inside python scripts to process raw strings and nested data dicts.

Leveraged Pandas and NumPy to cleanse game metadata, sorting system specs from unstructured catalog files.

Linux Server Environments



Configured file transfers and background procedures directly on cluster rigs utilizing secure SSH connections.

Utilized tmux terminal management tools to maintain uninterrupted task processing independent of connection drops.

Week 2: Advanced Server APIs & Integration

High-Performance Storage

Managed high-speed directories directly on high-performance local /raid storage systems.

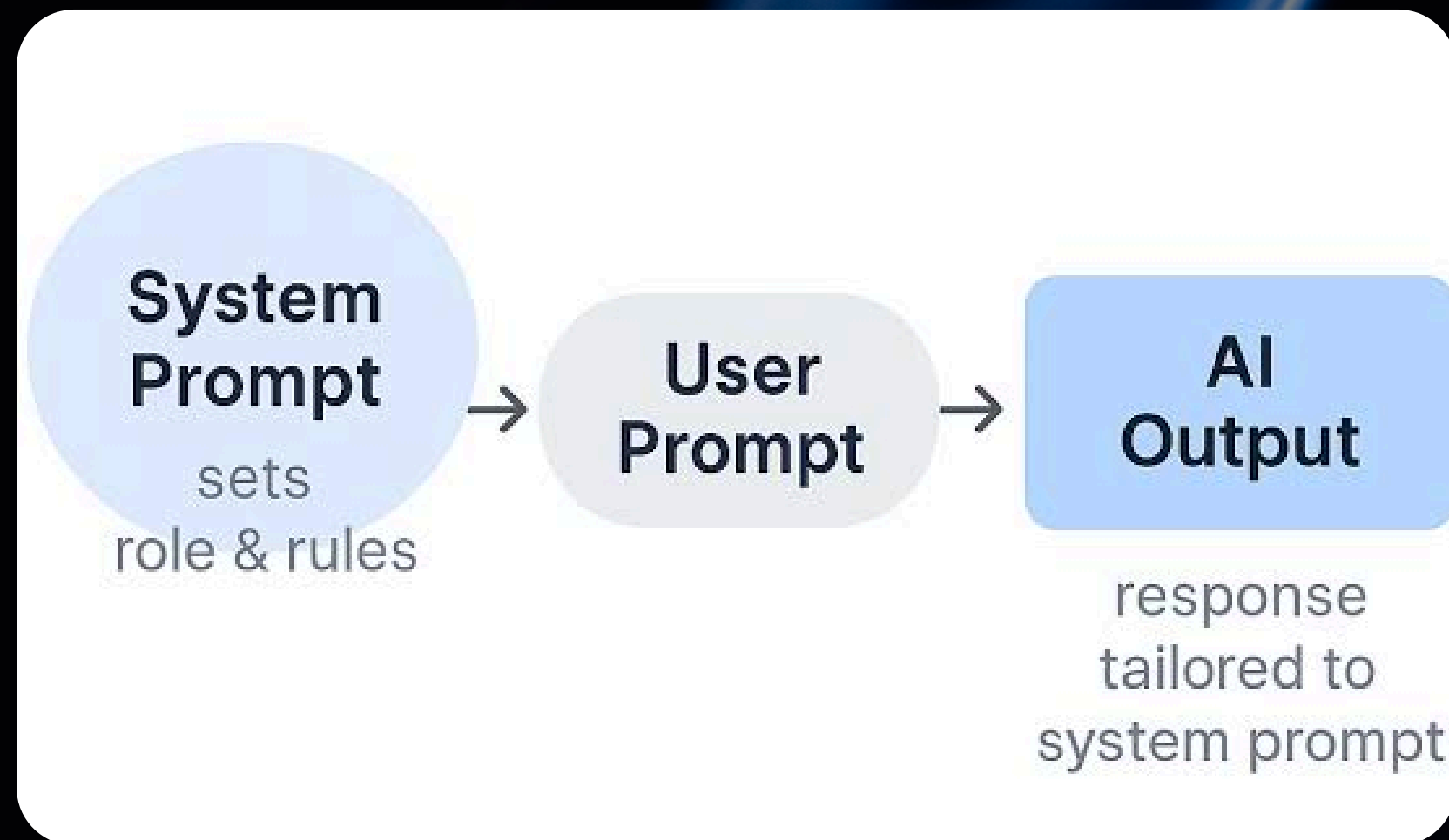
Monitored computation and graphic processing metrics dynamically using standard terminal nvidia-smi system logs.

Multi-API Stitching

Programmed asynchronous HTTP operations through standard python network requests packages.

Linked backend automatic speech-to-text (STT) layers to the central core LLM, finishing with voice generation (TTS).

Week 3: Advanced Prompting & Benchmarking



Prompt Engineering

Engineered robust custom instructional prompts containing explicit, locked context boundaries.

Formulated 5 core pillars in instruction design: Role, Context, Core Objective, Format constraints, and security guardrails.

Multi-Language Benchmark

Ran extensive evaluation benchmarks evaluating accuracy and formatting in Russian, Kazakh, and English.

Assessed system response latencies and verified generation quality through an objective 6-point evaluation scale.

Week 4: Frontend Development & UI Engineering

React State Flow

Crafted modular web application layouts utilizing reusable component methodologies.
Employed React useState hooks to dynamically refresh user visual states when receiving high-speed audio data streams.

Modern Interface Styling

Designed dark, clean, tech-inspired layouts styling CSS Flexbox frameworks directly on game browser interfaces.
Connected front-end microphone actions to processing back-end servers utilizing standard asynchronous JavaScript Fetch scripts.

Week 5: Vector DB Architecture & RAG

PHASE 1

Markdown Chunking

Wrote custom text parsing files to segment large gaming logs along structural line breaks (`\n\n`), avoiding loss of item context.

PHASE 2

Vector Encoding

Passed processed text snippets through local language tokenizers to build high-dimensional mathematical vector arrays representing core semantics.

PHASE 3

ChromaDB Queries

Integrated the open-source ****ChromaDB**** container locally, implementing fast cosine similarity lookups over all loaded data records

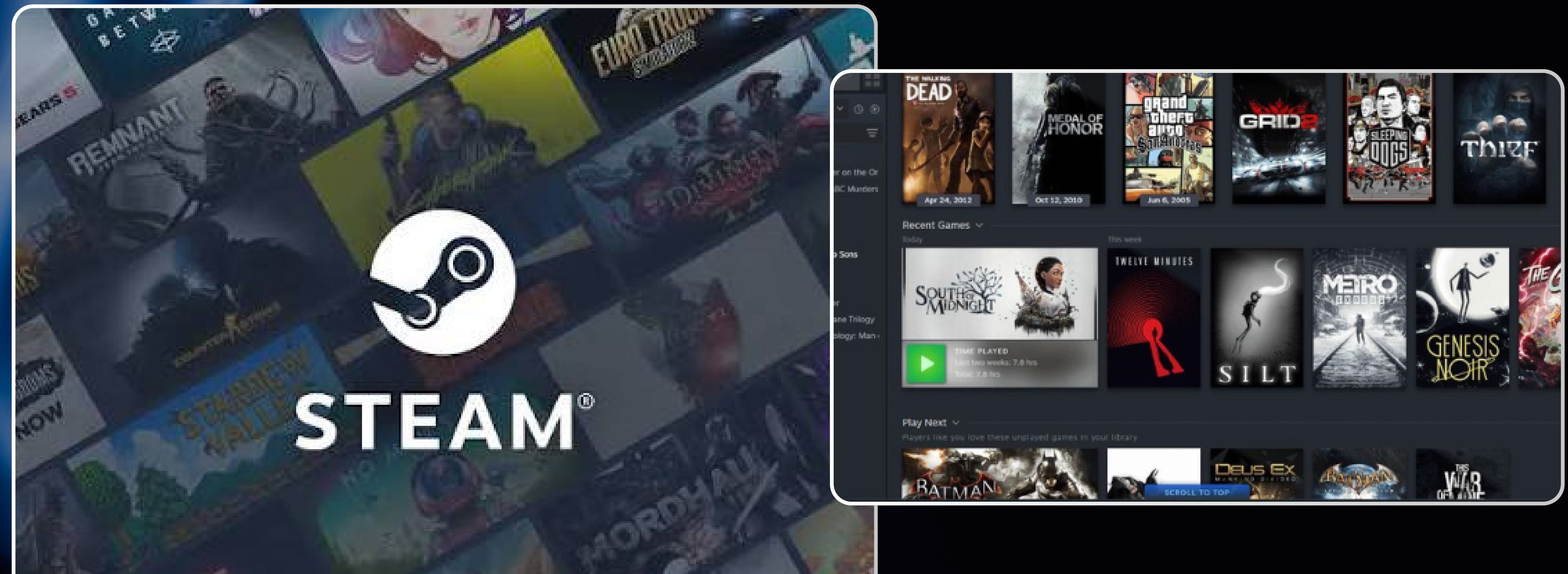
The Vision

Transforming the way we
find our next favourite game

We bridge the gap between abstract human emotions and rigid database tags. Our AI-driven 3D Avatar acts as an expert companion to navigate the immense digital catalog of the Steam universe.

For example: Our AI-driven 3D Avatar becomes a companion that understands this type of request.

A user can say: 'I want a game similar to GTA 5, but with a deeper story and better characters.'



The Paradox of choice

Library Fatigue: Over 100,000 titles on digital storefronts create immense choice paralysis for modern gamers.

Broken Metadata: Crowdsourced and community tags are frequently inaccurate, polluted, or highly misleading.

Commercial Bias: Storefront algorithms naturally favor massive marketing budgets over genuine indie quality.

The AI Solution: A local RAG system bypasses marketplace noise, delivering objective, context-anchored recommendations instantly.



Our Projects details

AI - STEAM GAME RECOMMENDATIONS

FIND YOUR NEXT GAME

Tell the AI what you're into — it searches a Steam game library and serves up titles picked for **you**. A 3D avatar walks you through every recommendation by voice, in real time.

RPG FPS Roguelike Co-op Indie Strategy Open-World Horror Souls-like

LANGUAGE

Kazakh

AVATAR

NEW Ava

ENVIRONMENT

NEW Gaming Room

VOICE

▶ START EXPERIENCE →

Connected

+ NEW CHAT

Search chats...

New Chat

Start a conversation

0 messages

ASSISTANT

AI Gaming Advisor

Powered by Groq + MangiSoz

STATUS

Knowledge DB ONLINE
Groq LLM ONLINE
MangiSoz TTS ONLINE
MangiSoz STT ONLINE

CONTROLS

VOICE Female
CAMERA Wide
LANGUAGE Kazakh

PIPELINE METRICS

Mic permission needed

Type your message...

Send →

Thank You For Attention

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